

Arnab Chakraborty

Full Stack Developer

📍 Guwahati, Assam

✉ that.zytact@gmail.com | 🌐 www.zytact.com | 🔗 linkedin.com/in/zytact | 🐙 github.com/zytact

Education

Gauhati University, Bachelor of Technology (B.Tech.) in Information Technology Aug 2023 - May 2027
(Expected)

Relevant Coursework: Computer Science fundamentals, Data Structures, OOP, DBMS, Operating Systems

Experience

Full Stack Developer Intern, FRINT Mar 2026 - Apr 2026

- Built an enterprise-grade AI proctoring system with React + Vite frontend and Express/Socket.io gateway.
- Developed Python FastAPI microservices for AI/ML using Whisper STT, MediaPipe, OpenCV, MongoDB, Redis, Groq, and Gemini APIs.
- **Technologies:** React.js, TypeScript, Python, FastAPI, Node.js, Express, Socket.io, MongoDB, Redis, WebRTC, Groq, Gemini, Whisper, MediaPipe, OpenCV

Research Intern, National Institute of Technology, Silchar Jun 2026 - Jul 2026

- Built a multimodal deep learning pipeline under Dr. Ram Kumar Karsh on the RASTA retinal OCTA dataset, fusing three retinal layers, clinical data, and U-Net++ segmentation masks through SimCLR-pretrained ConvNeXt encoders and cross-layer attention.
- Reached 0.80 patient-level accuracy and 0.898 macro ROC-AUC under leakage-safe patient-grouped 5-fold cross-validation, with temperature calibration, MC-dropout uncertainty, and Grad-CAM/SHAP explainability.
- **Technologies:** Python, PyTorch, ConvNeXt, U-Net++, SimCLR, OpenCV, scikit-learn, SHAP

Open-Source Contributor | MetaBrainz Picard Nov 2024 - Dec 2024

- Implemented a feature to show old and new file paths during renaming, reducing user confusion by 30%. (PR #2563)
- Built a custom title-casing function fixing 100+ capitalization issues and improving metadata consistency; added Pytest unit tests increasing coverage. (PR #2548)
- **Technologies:** Git, GitHub, Python, PyQt, Pytest

Projects

InvenTrack - 2025 inventrack.zytact.com

- Built a multi-tenant inventory platform with per-tenant data isolation, automated reorder triggers, and dynamic pricing rules in a single PostgreSQL schema.
- Integrated Gemini LLM to generate demand forecasts and restock recommendations from live inventory data, requiring careful prompt design to produce reliable, structured output.
- **Technologies:** Next.js, Clerk, TypeScript, Tailwind CSS, Shadcn UI, TanStack Query, Drizzle ORM, Gemini API

Compress - 2026 compress.zytact.com

- Built a privacy-first image compressor using Rust/WASM with Lanczos resampling and target-size binary search; 100% client-side, no server uploads or tracking.
- **Technologies:** Rust, WebAssembly, TypeScript

Strokeshare - 2025 strokeshare.zytact.com

- Built an infinite-canvas drawing app with real-time shape rendering; engineered precise eraser hit-detection that works correctly at any zoom level using coordinate-space normalization.
- Implemented undo/redo as full-state snapshots with branch erasure on new actions; handled transform decomposition for stable serialized geometry after resize/rotate.
- **Technologies:** Next.js, TypeScript, Tailwind CSS, React Konva, Zustand

Technical Skills

Programming Languages: Python, C, JavaScript, TypeScript, HTML, CSS

AI / Machine Learning: Generative AI, Large Language Models (LLMs), Prompt Engineering, PyTorch, Computer Vision, Hugging Face Transformers, OpenAI API, Gemini API, NumPy, Pandas, scikit-learn

Front End Web: React.js, Next.js, Tailwind CSS, TanStack Query, Zustand, React Konva, Web design

Back End: Node.js, Express.js, Python, RESTful APIs, tRPC, Auth.js, Better Auth, Clerk

Database: MongoDB, PostgreSQL, MySQL **Exposure:** AngularJS, PHP, .NET, Ruby, Oracle

Tools: Git, GitHub Actions, Linux, Vim/NeoVim, Jest, Vitest, Pytest, ESLint, Prettier